

KIRAN SUKUMARAN

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 Thiruvananthapuram, Kerala, India

PROFESSIONAL SUMMARY

Technical Lead and Backend & AI Engineer with 8+ years of experience building scalable backend systems, cloud-native applications, and AI-powered solutions. Strong expertise in Node.js, TypeScript, NestJS, microservices, and AWS, with recent focus on Generative AI and Voice AI. Experienced in architecting real-time Voice Agent pipelines integrating Speech-to-Text (STT), Large Language Models (LLMs), and Text-to-Speech (TTS), while leading engineering teams and delivering enterprise solutions across healthcare, marketing, ecommerce, cybersecurity, and workforce management.

CORE TECHNICAL SKILLS

- Languages: TypeScript, JavaScript, Dart, Java, Kotlin
 - Backend & Architecture: Node.js, NestJS, REST APIs, Microservices, Serverless, Distributed Systems
 - AI & GenAI: Generative AI, LLMs, AI Agents, Voice AI, STT, TTS, Prompt Engineering
 - Cloud & DevOps: AWS (Lambda, EC2, S3, RDS, CloudWatch), GCP, Docker, Pulumi, Infrastructure as Code
 - Databases: PostgreSQL, MongoDB, MySQL
 - Messaging: Kafka, RabbitMQ, MQTT
 - Frontend & Mobile: Angular, Next.js, Flutter, Three.js, WebGL
 - Tools: Git, GitHub, Postman, Jira, VS Code
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EXPERIENCE

Associate Technical Lead

KeyValue Software Systems | *Jan 2026 – Present* | Thiruvananthapuram, India

- Designed and deployed end-to-end Voice AI Agent pipelines integrating Speech-to-Text (STT), LLM orchestration, and Text-to-Speech (TTS) for real-time conversational applications.

- Designed and optimized low-latency voice workflows using Deepgram and Cartesia.
- Engineered Generative AI workflows for image and video generation.
- Designed scalable AWS infrastructure using Pulumi and Infrastructure as Code (IaC).
- Built highly available backend services and serverless architectures using AWS Lambda.

Lead Engineer

QBBurst Technologies | Sep 2020 – Jan 2026 | Thiruvananthapuram, India

- Led and mentored a 10+ member backend engineering team delivering microservice-based applications using Node.js, TypeScript, and NestJS.
- Designed backend solutions across healthcare, marketing, and ecommerce, including data, analytics, and automation workflows.
- Improved project delivery efficiency by 20% and optimized resource allocation, contributing to a 15% reduction in project costs.
- Developed marketing automation and content distribution solutions that reduced content creation time by 50% and increased reach and engagement by 75%.
- Built and maintained AWS applications using EC2, Lambda, RDS, and CloudWatch, including systems handling sensitive healthcare data.
- Mentored engineers, conducted code reviews, contributed to technical planning, and collaborated with clients and stakeholders.

Associate Software Engineer

Pivot Systems Inc. | Jun 2018 – Sep 2020 | Thiruvananthapuram, India

- Designed and delivered a 3D infrastructure visualization and planning solution using Angular, Three.js, WebGL, Node.js, and MongoDB.
- Developed secure REST APIs and web gateways with role-based access control.
- Developed staff management, attendance, wellness, leave, and work schedule management systems.
- Supported cloud deployments, monitoring, and application performance optimization.

Independent / Freelance Software Engineer

Freelance | 2015 – Present

- Developed customized software solutions for businesses and clinics across India, Singapore, and Indonesia.
- Built web-based booking, inventory, client management, and business workflow applications using Node.js, Flutter, and Firebase.

- Worked directly with clients across requirements gathering, architecture, development, deployment, and production support.
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SELECTED PROJECTS

1. Agentic Voice AI & Generative AI Platform

Tech: Node.js, TypeScript, AWS, Deepgram, Cartesia, LLMs, STT, TTS

- Built end-to-end agentic voice pipelines combining STT, LLM reasoning and orchestration, TTS, and real-time backend processing.
- Developed GenAI workflows for image and video generation with scalable cloud infrastructure and backend orchestration.

2. Bayer Global – Healthcare Engagement Platform

Tech: Next.js, Node.js, PostgreSQL, Prisma

- Developed backend capabilities for healthcare engagement, surveys, newsletters, journey management, and performance tracking.
- Supported data-driven marketing and analytics workflows for healthcare professionals and medical representatives.

3. Ecommerce & Marketing Platforms

Tech: Node.js, TypeScript, NestJS, AWS, PostgreSQL, MongoDB

- Developed scalable backend services for ecommerce, marketing automation, content distribution, inventory, and customer workflows.
- Built automation and analytics capabilities to improve content delivery and campaign effectiveness.

4. Tenable – Security Platform Modernization

Tech: Kotlin, Java, PostgreSQL, Node.js

- Contributed to migrating legacy NASL services to Kotlin-based microservices.
- Developed Node.js controllers and contributed to secure service integrations.

5. Hitachi – 3D EZ Design Simulator

Tech: Angular, Node.js, WebGL, MongoDB, Three.js

- Developed 3D simulation capabilities for camera and sensor placement, infrastructure measurements, and security-system planning.
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COMMUNITY EXPERIENCE

Backend Developer / Community Contributor
Prathidhwani Technical Forum | *Jun 2020 – Apr 2022*

- Contributed backend development and technical implementation for community-focused platforms supporting job seekers and professionals.
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EDUCATION

B.Tech – Computer Science and Engineering
Valia Koonambaikulathamma College of Engineering & Technology
Kerala University | 2018

CERTIFICATIONS

Generative AI Foundations
Generative AI – Fundamentals
Microsoft Azure Fundamentals
Full Stack Development – Mashup Stack

 <https://www.mashupstack.com>